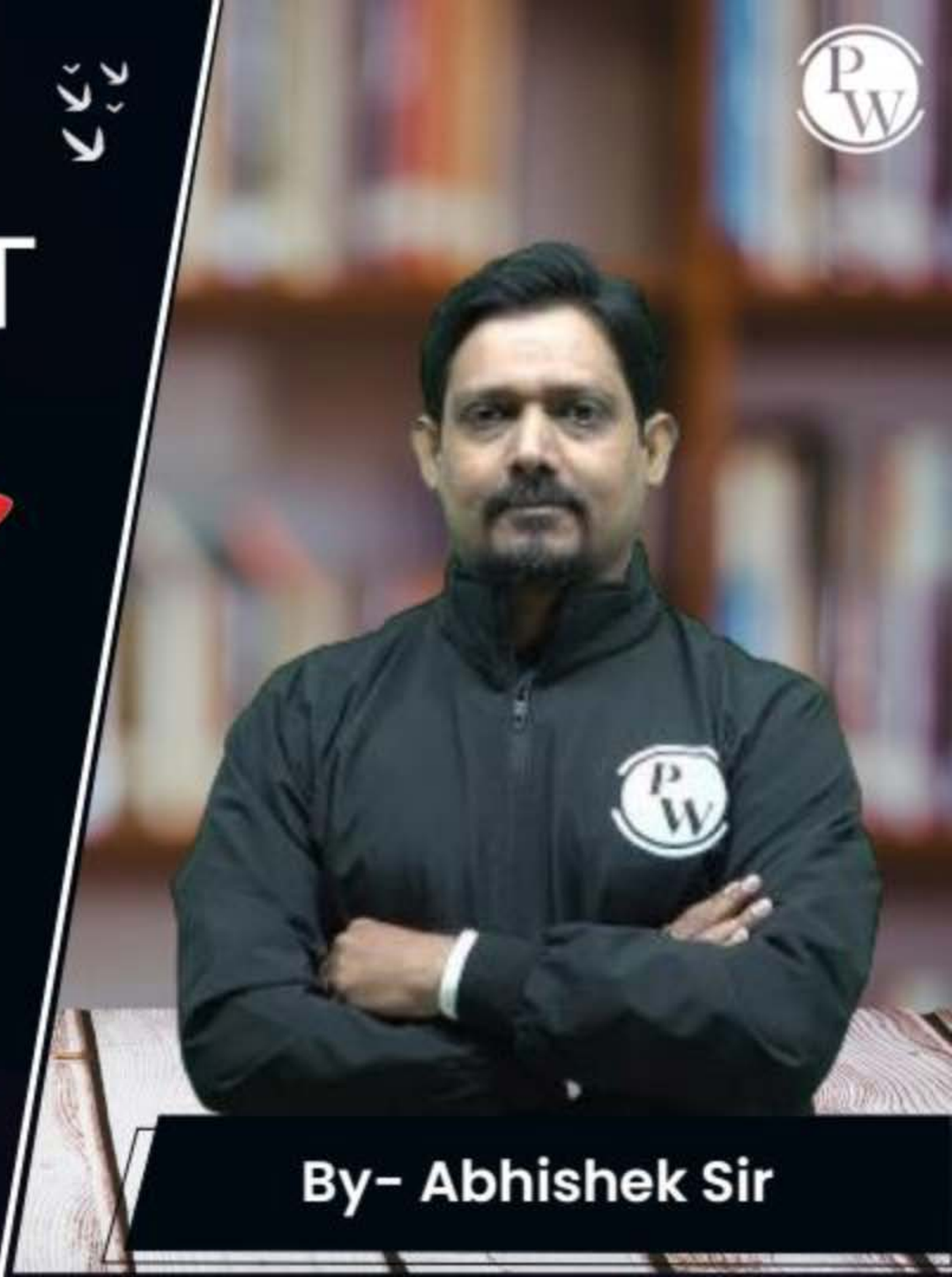
Computer Science & IT



Fundamental of C Language

Functon & Storage Class

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic: do-while
- Topic: while Loop
- Topic: break
- Topic: Continue
- Topic

Slide

Topics to be Covered



- Topic — function
- Topic — declaration & definition
- Topic — function parameter passing Mechanism
- Topic
- Topic

Slide



Function

```
int main(){
```

← Main function

```
}
```



Function

What is a function :. function is Sequence of instruction that forms Single logical unit.

A {

S_1;
S_2;
S_3;
S_4;

}

Single unit

Slide



Function

Why function Required :- C is structured programming Language

Structure means Blocks

e.g. update & print ← 10 times call in program

Suppose cocle size is 5k

10 x 5 = 50k

Slide



Function

update & print make function

and wherever required simply call the function

function Call

Slide



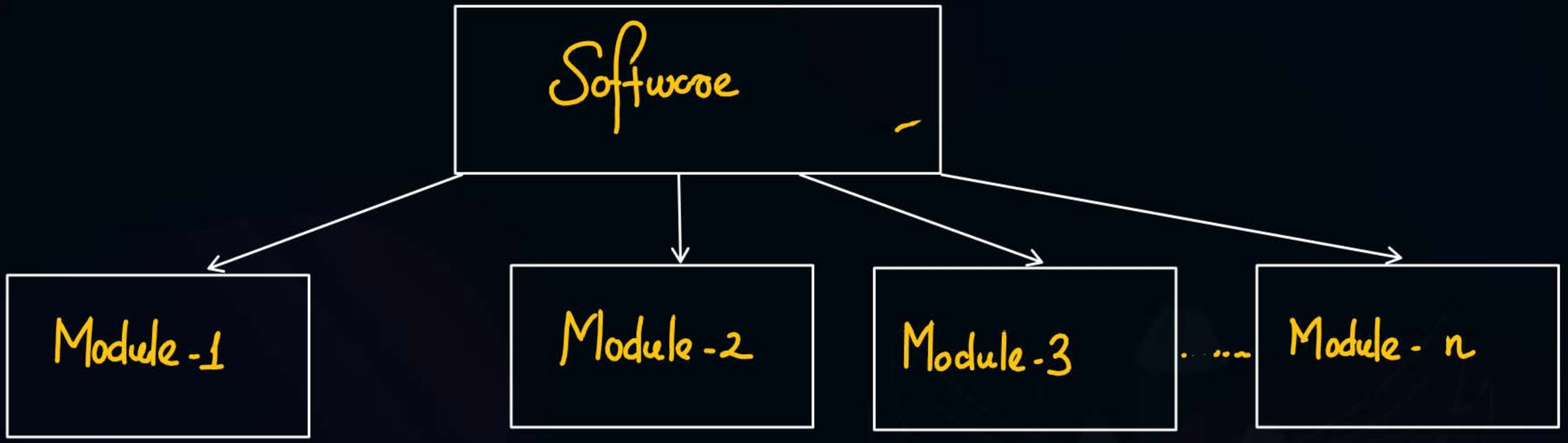
Function

1. Management of code & reusability

2. Modularity of program



Function





Function

Modularity is breaking the program into Smaller parts.

for each Module we write function



Function

```
# include<stdio.h>
int add ( int, int);
```

function prototype : Attribute of function

1 Return type (int)

2 Name of function : add

3 parameter of function (2 integer parameter)

Slide



Function

```
# include<stdio.h>
int add(int,int);
int main(){
    int a=20, b=30,
    int c = add(a,b);
    printf("%d", c);
    return 0;
    }
```

→ forward declaration

function Call

Slide



Function

```
# include<stdio.h>
int add ( int, int);
int main() {
    int a=20, b=30,
    int c = add(a.b);
    printf ("%d", c),
    return 0,
    }
```

function definition

```
int add ( int c, int d){
        int Sum,
        sum= c+d,
        return sum;
    }
```



Function

```
# include<stdio.h>
int add (int, int);
int main() {
    int a=20, b=30,
                    50
    int c = add(a, b);
    printf("%d", c); //50
    return 0;
    }
```

function definition

```
c=a  c=20   d=6  d 30
int add ( int c, int d){
    int Sum,
    sum= c+d, // Sum: 20+30=50
    return Sum; //return 50
}
```

Slide



Function

1 When function is called control transfer to body of function.

2. a, b is called actual parameter
   c, d is formal parameter

3 Actual parameter value copied to formal parameter

c:a, d:b

Call by value

Slide



Function

4. function body executes and then if function is returning a value then value return.
5. control transfer back to callee function (Main function)
6. Rest of code executes



Function

1 Is forward declaration required? – No

2 What if No return value & No parameter passed :- ok



Function

```
# include<stdio.h>
void fun( ) {
    printf(":Name");
    return;
}
int main() {
    fun(); // printf("Name");
    return 0;
}
```

← function definition

void : No return value

( ) parameter is blank. No parameter

Slide



2 mins Summary

Topic function

Topic function declaration

Topic function definition

Topic function execution & purpose

Topic

Slide

THANK - YOU

